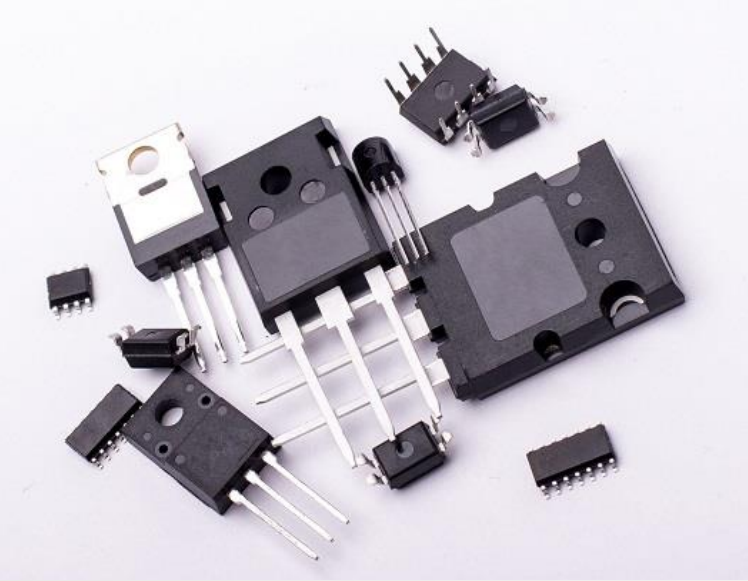
Electronică Analogică

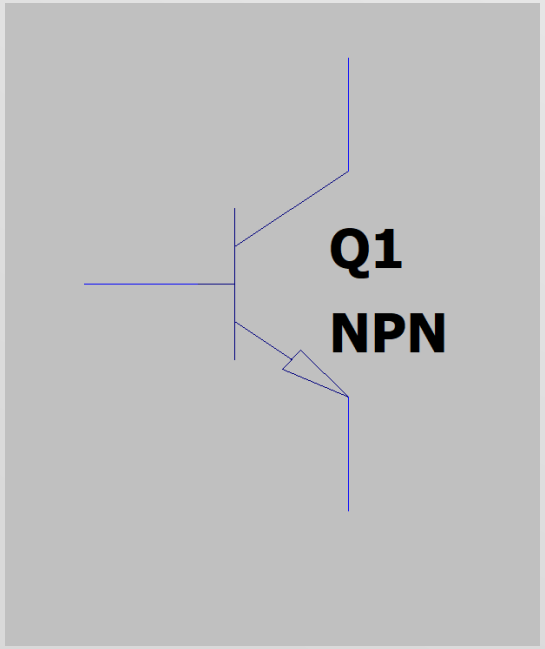


© https://www.questcomp.com/blog/transistors-the-key-to-modern-electronics

Disclaimer

- *Slide-urile acestea reprezintă un sport de curs / rezumat.*
- *Slide-urile nu acoperă toate explicațiile și demonstrațiile oferite în timpul cursului.*
- *Slide-urile nu limitează în vreun fel materia cerută la examen.*
- *Pentru varianta completă, sunteți așteptați la curs*

TBIP

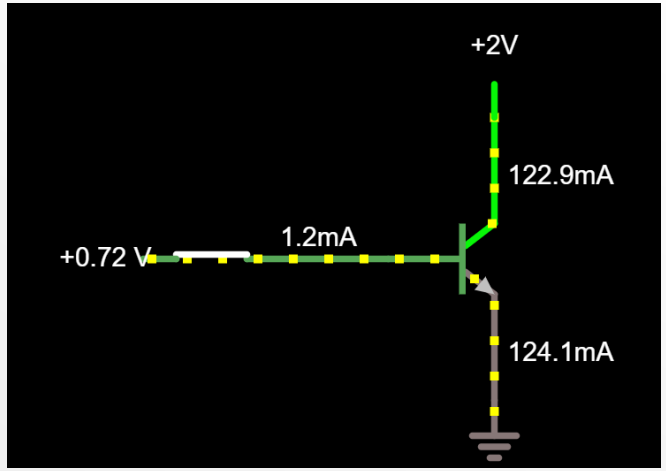


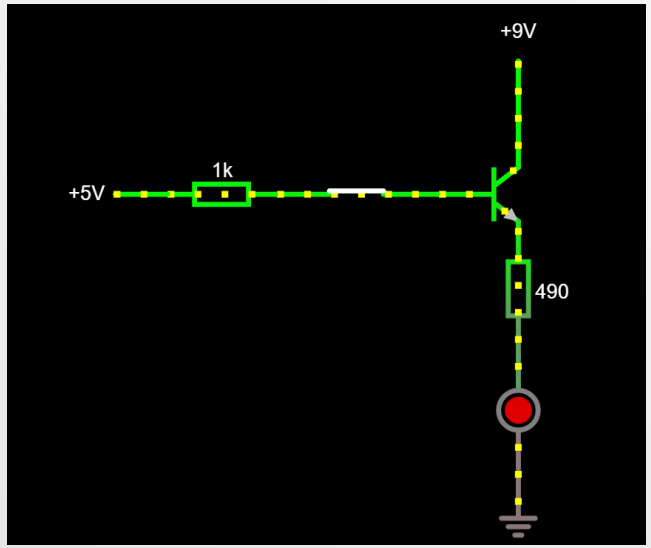
https://tinyurl.com/y2ydzdmj

Explicatii generale

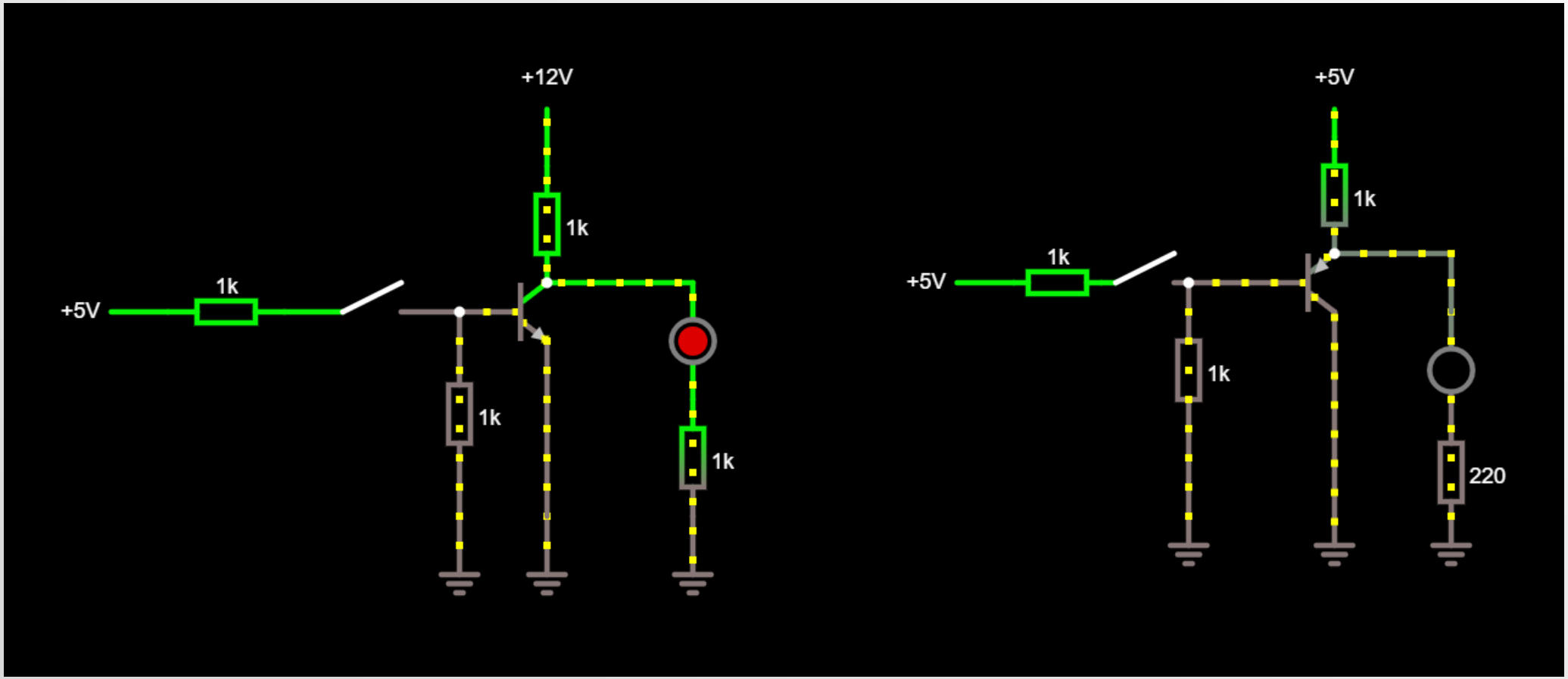
Utilizari https://tinyurl.com/y2e76j62

Exemple





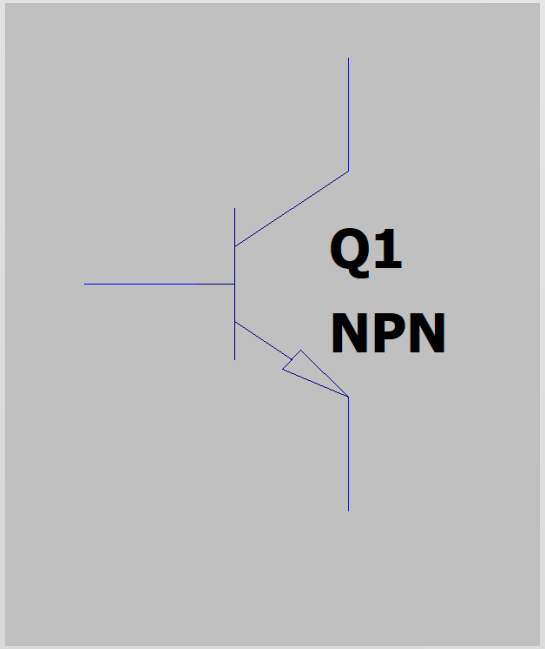
Exemple de utilizare TBIP (introductive)

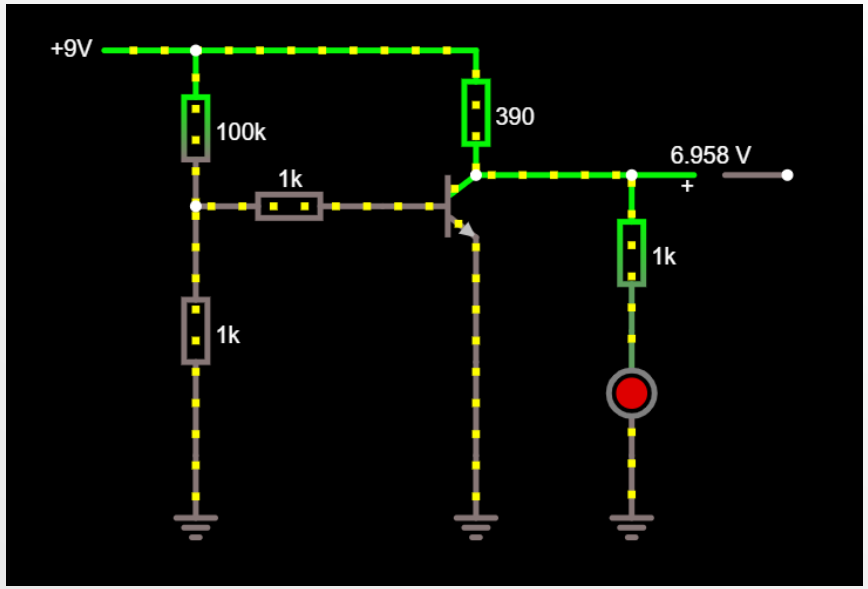


https://tinyurl.com/y5tgl8ct

Exemple de utilizare TBIP (introductive)

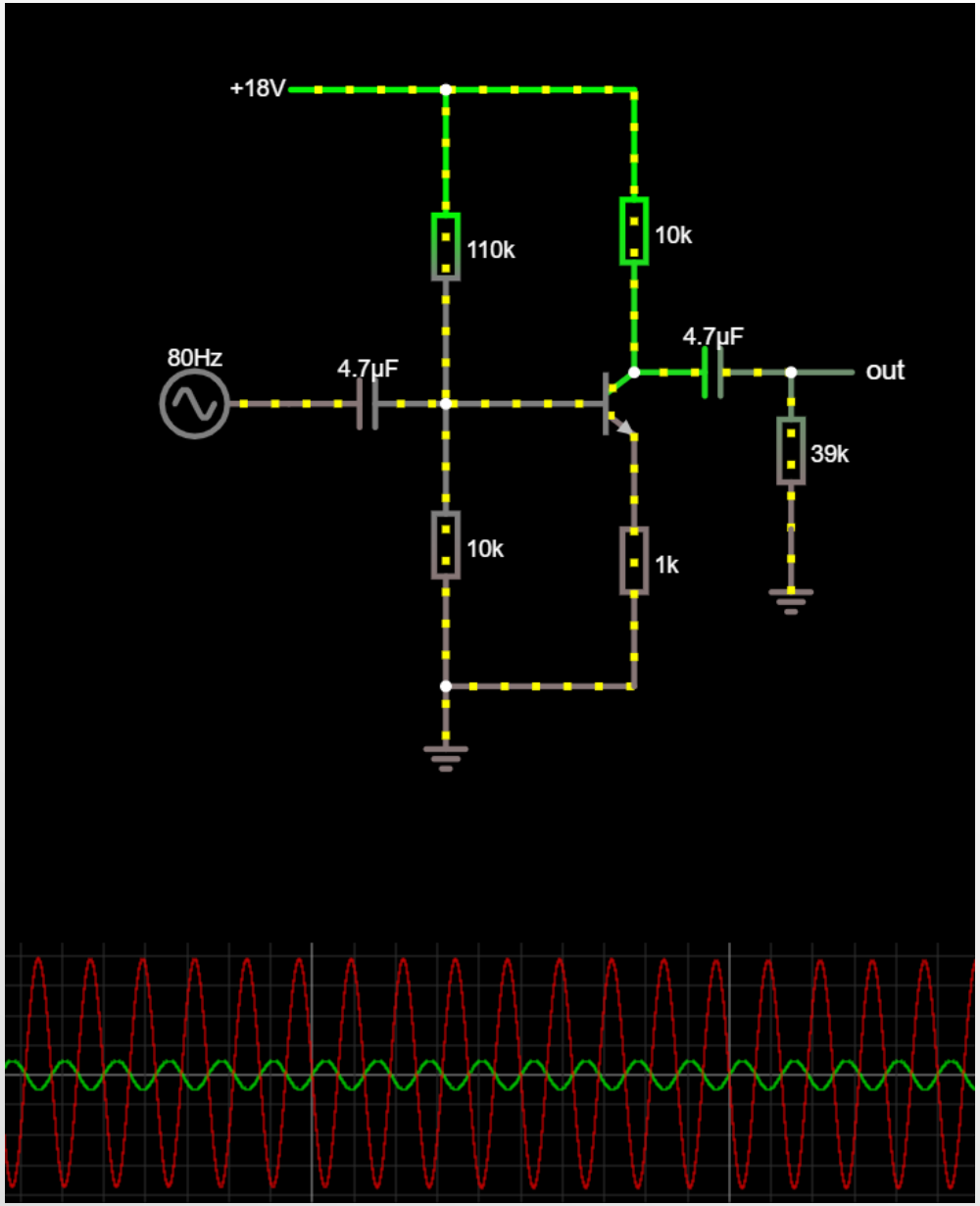
https://tinyurl.com/yymwm76u



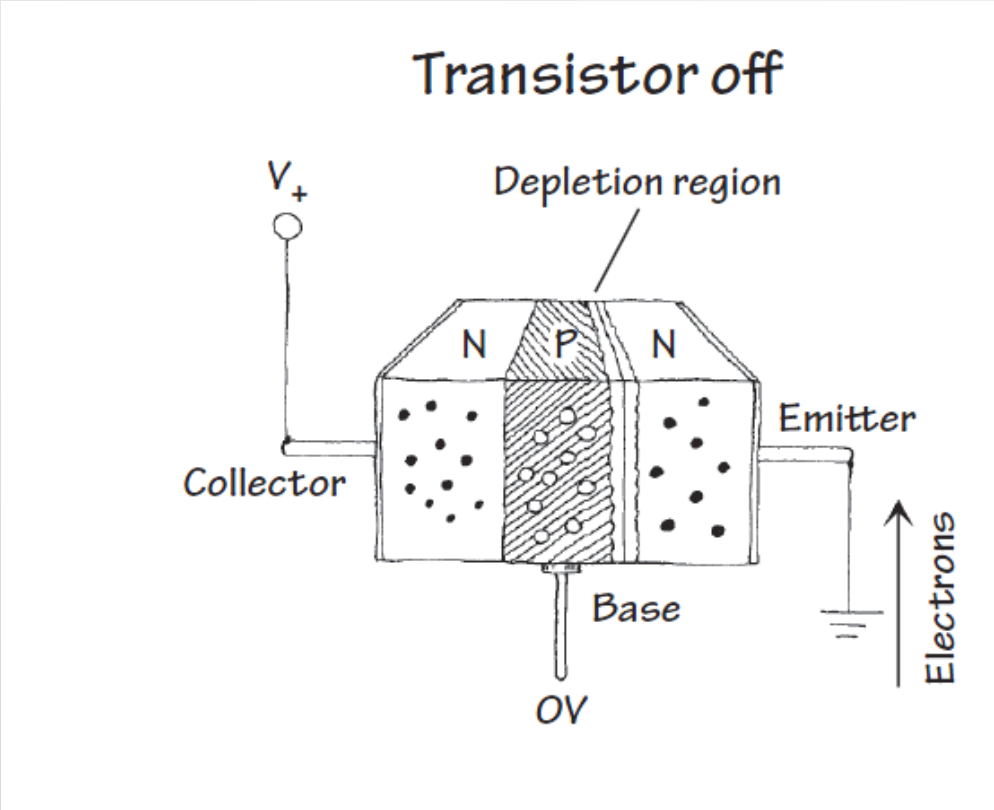


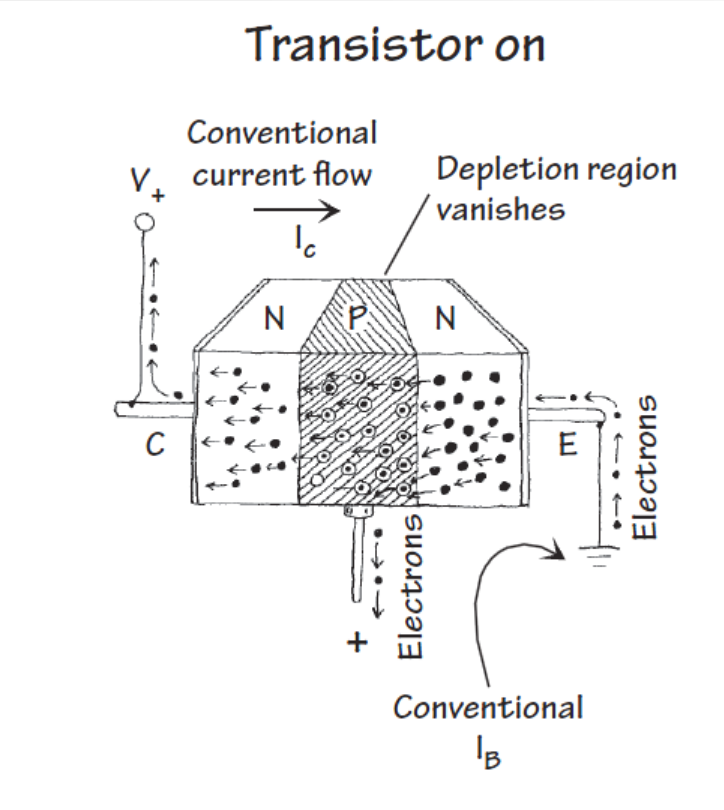
Exemple de utilizare TBIP (introductive)

https://tinyurl.com/yy6yfrw6



TBIP

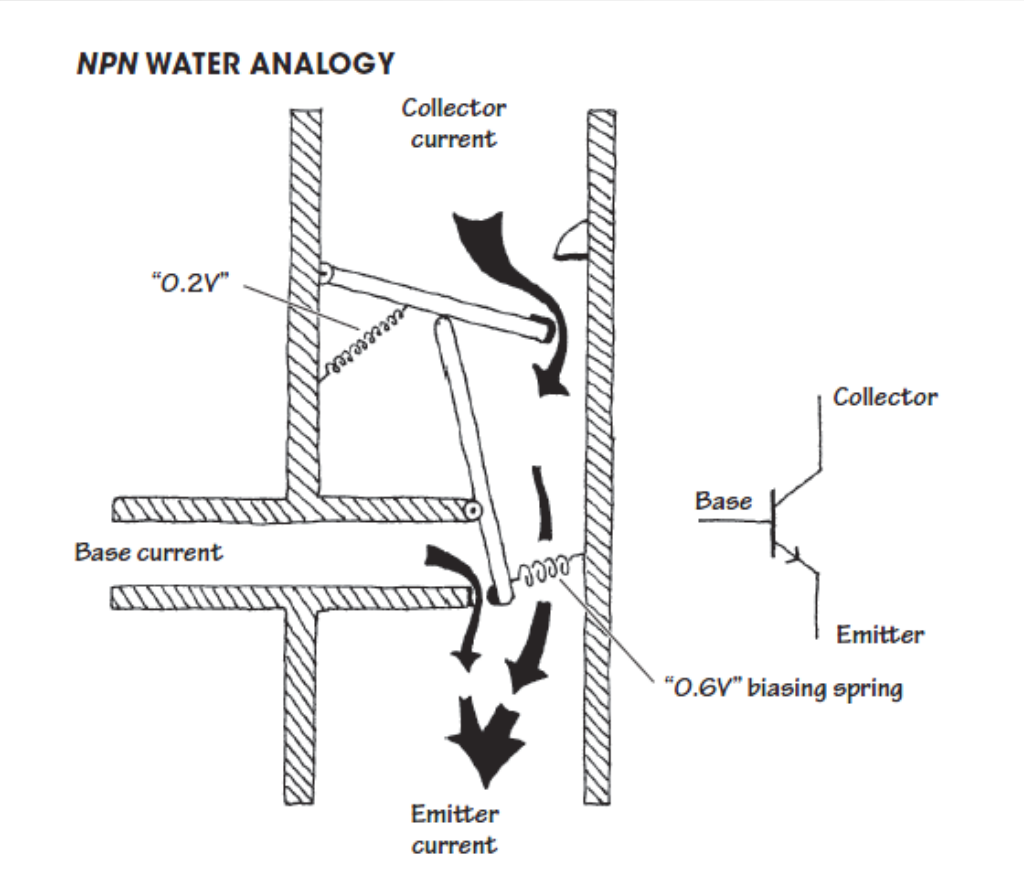




Explicatii la curs

©Practical Electronics for Inventors (Paul Scherz și Simon Monk, 2016) - imagini

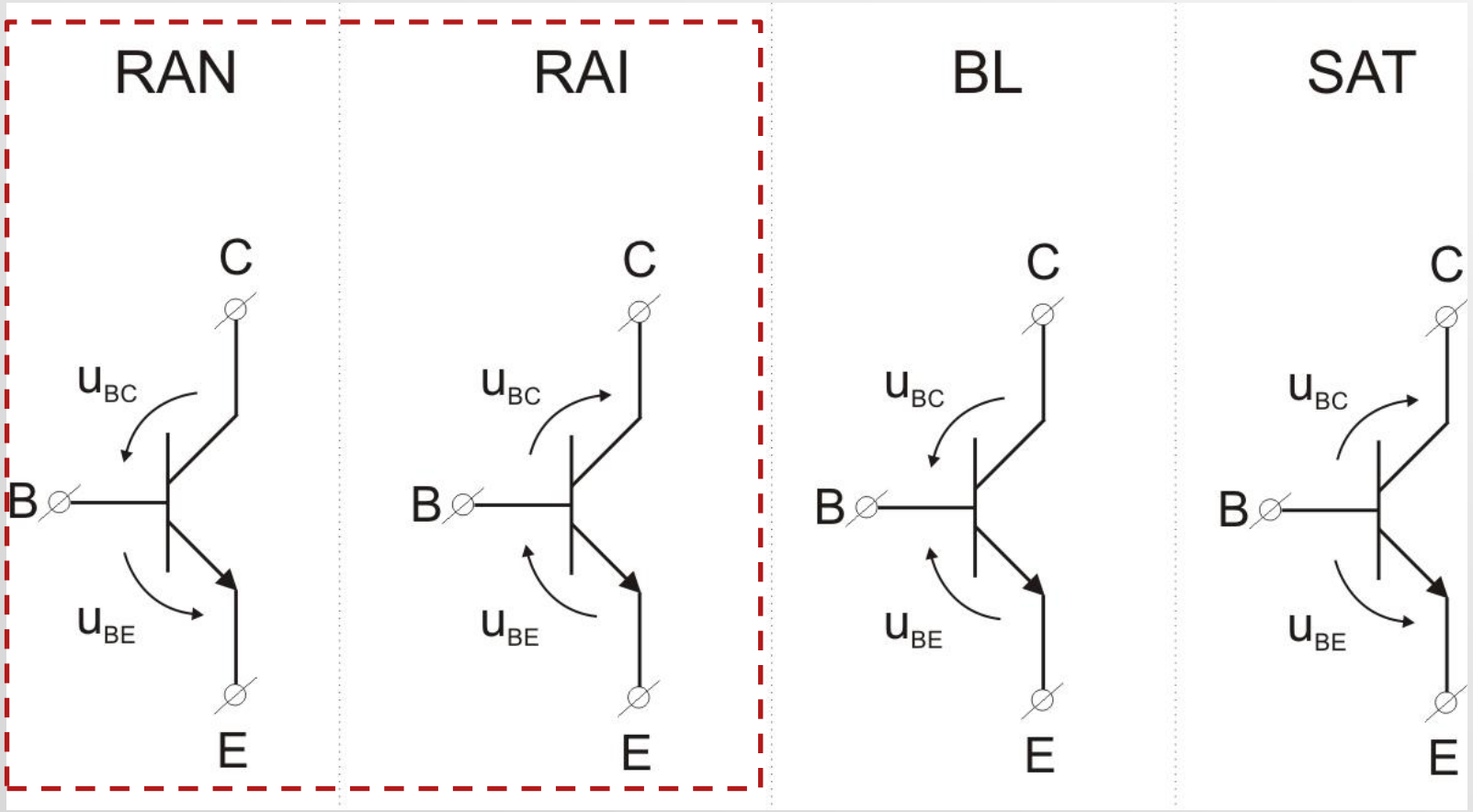
TBIP



©Practical Electronics for Inventors (Paul Scherz și Simon Monk, 2016) - imagini

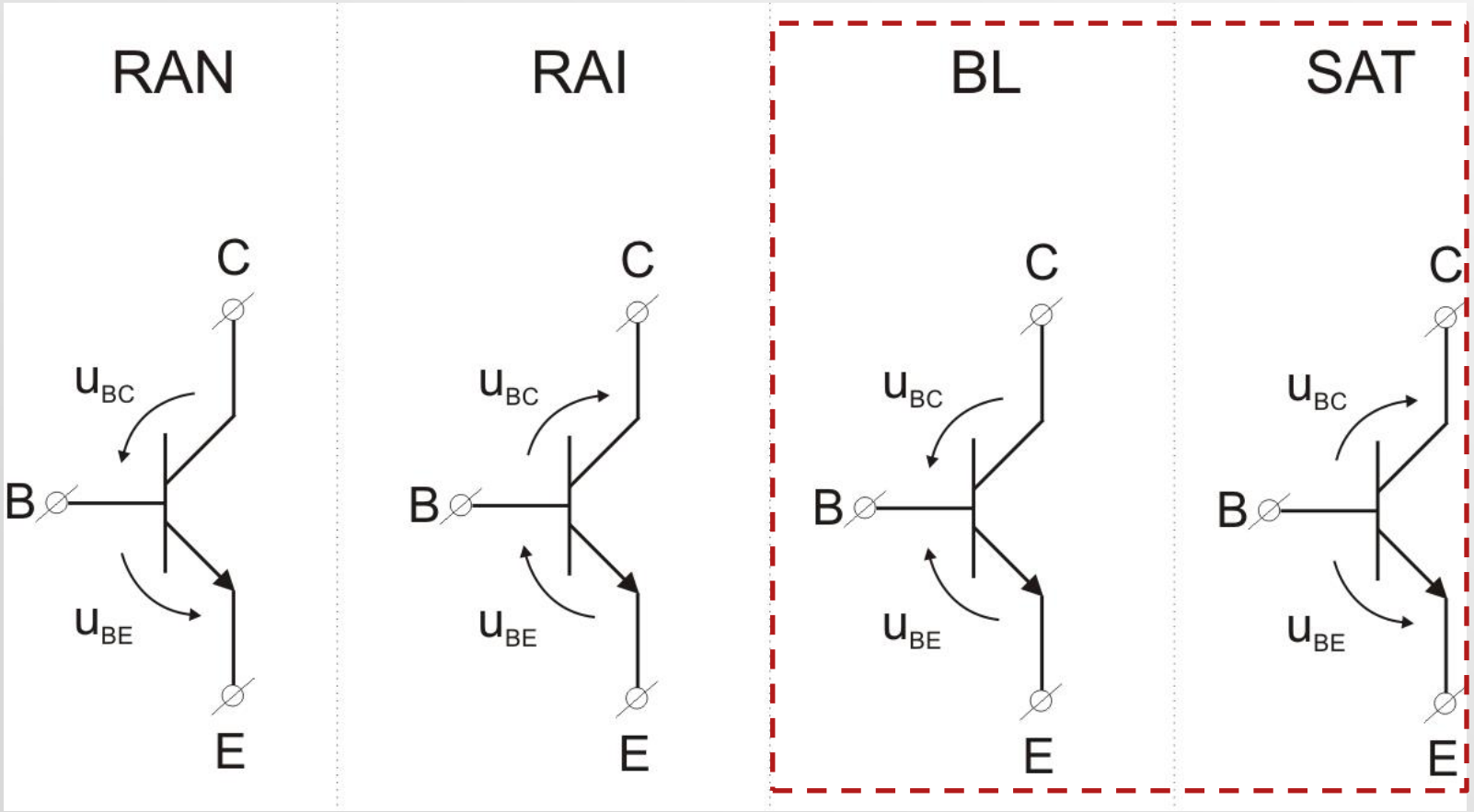
Polarizarea joncțiunior

Regimurile de lucru



$$i_c = \beta i_b + I_{ce_o}$$

Regimurile de lucru



Calculare PSF

PSF

- **Punctul Static de Funcționare** – I_C, I_E, I_B, U_{BE}, U_{CE}, U_{CB}
- Se calculează pentru un circuit în regim static
  - se ține cont doar de sursele de curent continuu
  - sursele de semnal se pasivizează (se pun la masă)
  - traseele din circuit ce conțin un condensator se elimină (regim static = curent continuu, după stabilizarea circuitului => prin condensatoare nu circulă curent)
- Pentru fiecare tranzistor dintr-o schemă se determină:
  - **I_C** și **U_{CE}** (din I_C și U_{CE} se deduc usor I_E, I_B, U_{BE}, U_{CB})
- Se dau: β, U_{BE}

PSF – pași de rezolvare

1. Se redesenează schema echivalentă în regim static
2. Se calculează curenții I_c:
    1. Se caută bucle / trasee în circuit dealungul cărora se poate scrie KII și există o singură necunoscută. În general, acestea cuprind joncțiunile BE ale tranzistorilor (U_{BE} se știe)
    2. Dacă nu se găsește o astfel de buclă/traseu, se caută puncte în care se poate calcula tensiunea folosind Thevenin sau divizor de tensiune.
    3. Se revine la punctul 2.1.
3. Se calculează tensiunile U_{CE}.
    1. Se folosește KII. Uneori este facilă calcularea U_{CB} și calcularea U_{CE} pe baza U_{CB} și U_{BE}.

PSF - Exemplu 1

$\beta = 100$

$U_{BE} = 0,6V$

PAS 1: calculăm I_{c1}

KII(1): $R_{b1} * I_{b1} + U_{BE} = V_{cc}$

$R_{b1} * I_{b1} + 0,6V = 6V$

$540k * I_{b1} = 5,4V \Rightarrow I_{b1} = 0,01mA$

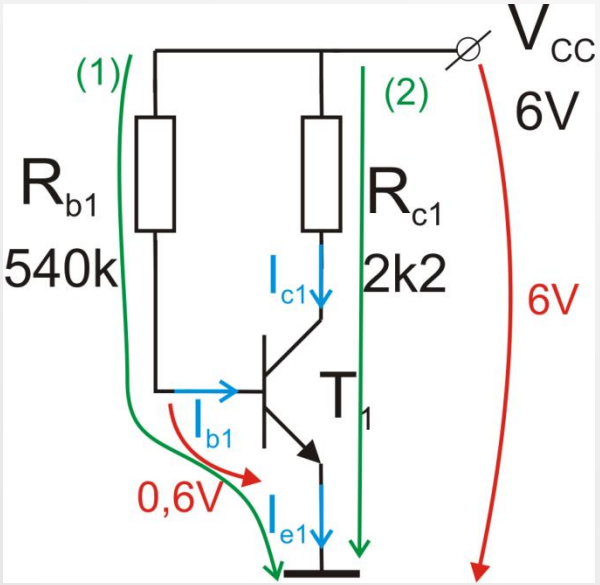
Cum $I_{c1} = I_{b1} * \beta \Rightarrow I_{c1} = 1mA$

PAS 2: calculăm U_{ce}

KII(2):

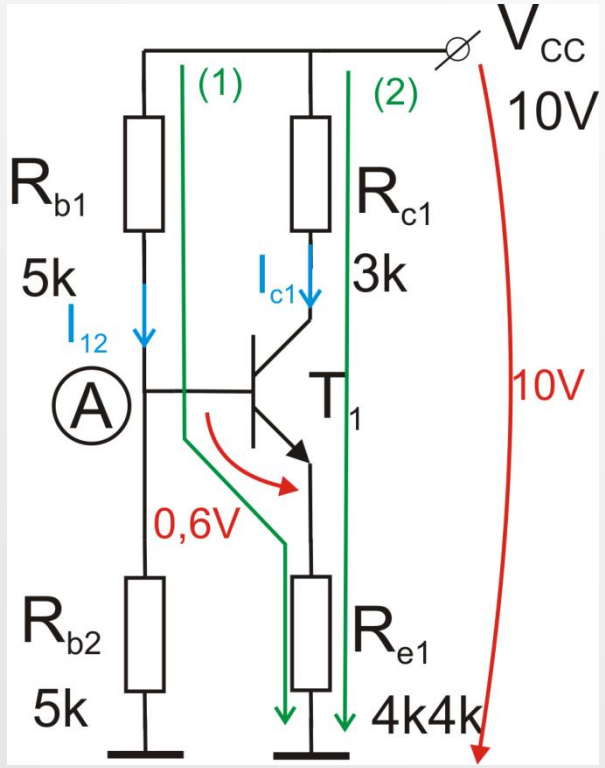
$R_{c1} * I_{c1} + U_{CE} = Vcc$

$R_{c1} * I_{c1} + U_{CE} = 6V \Rightarrow 2,2V + U_{CE} = 6V \Rightarrow U_{CE} = 3,8V$



PSF – Exemplu 2

$\beta = 100$
$U_{BE} = 0,6V$



PAS 1: calculăm I_{c1}

- Varianta 1:

Cum I_b f mic comparativ cu I_{12}, putem considera I prin R_{b1} = I prin R_{b2} => I_{12} = 1mA

Din **KII (1)** reiese:

$R_{b1} * I_{12} + U_{BE} + R_{e1} * I_{e1} = 5V$

$5V + 0,6V + 4k4\Omega * I_{e1} = 5V => I_{e1} = 1mA = I_{c1}$

- Varianta 2:

$V(A) = V_{cc} * R_{b2} / (R_{b1} + R_{b2}) = 5V$ (div tensiune)

Din **KII (1)** reiese: $0,6V + 4k4\Omega * I_{e1} = 5V => I_{c1} = 1mA$ *(Obs: dacă aplicăm Thevenin, $V_e = 5V$, $R_e = 2k5$; cum I_b si R_e sunt mici, se neglijează căderea de tensiune de pe R_e)*

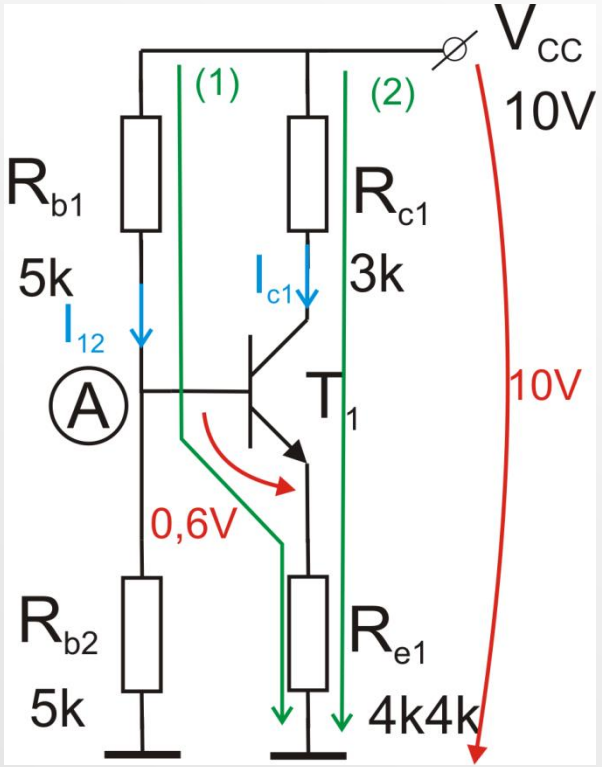
PSF – Exemplu 2

PAS 2: calculăm U_{CE}

Din KII (2):

$R_{c1} * I_{c1} + U_{CE} + R_{e1} * I_{e1} = 10V$

$3V + U_{CE} + 4,4V = 10V \Rightarrow U_{CE} = 1,7V$



Ex sim

https://tinyurl.com/5ykvzukr

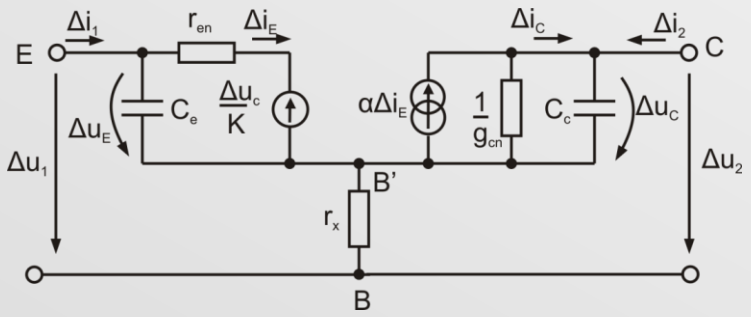
https://tinyurl.com/4zb2szxk

Modele clasice

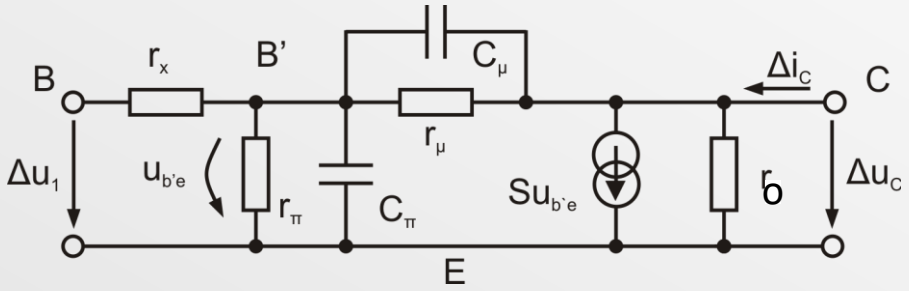
Semnale variabile mici;

Circuitul de intrare

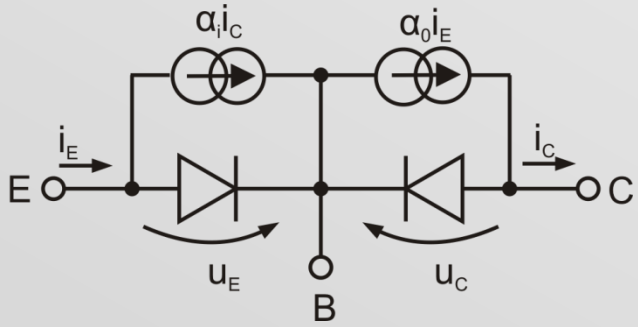
Early



Giacoletto

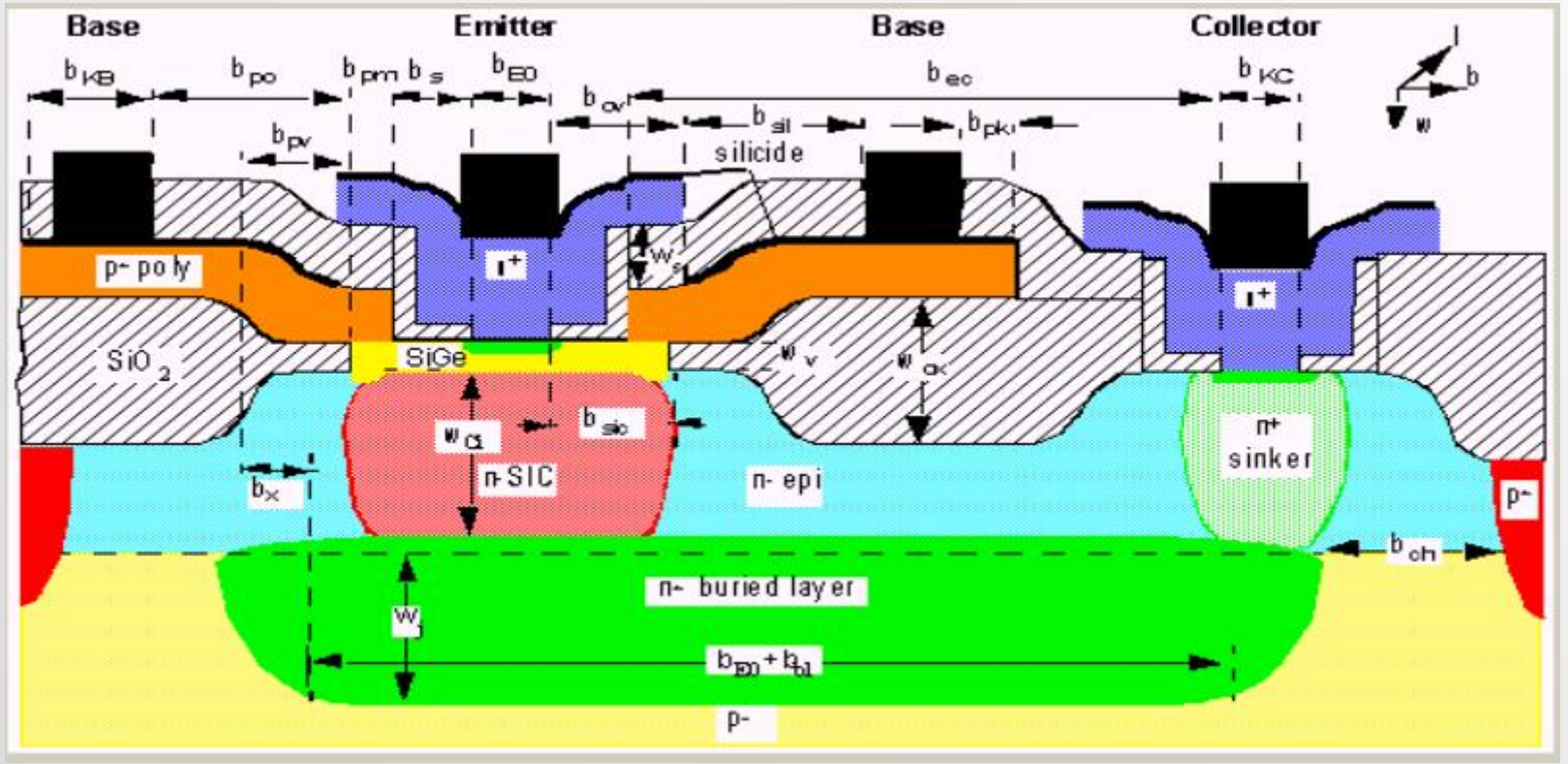


Ebers-Moll



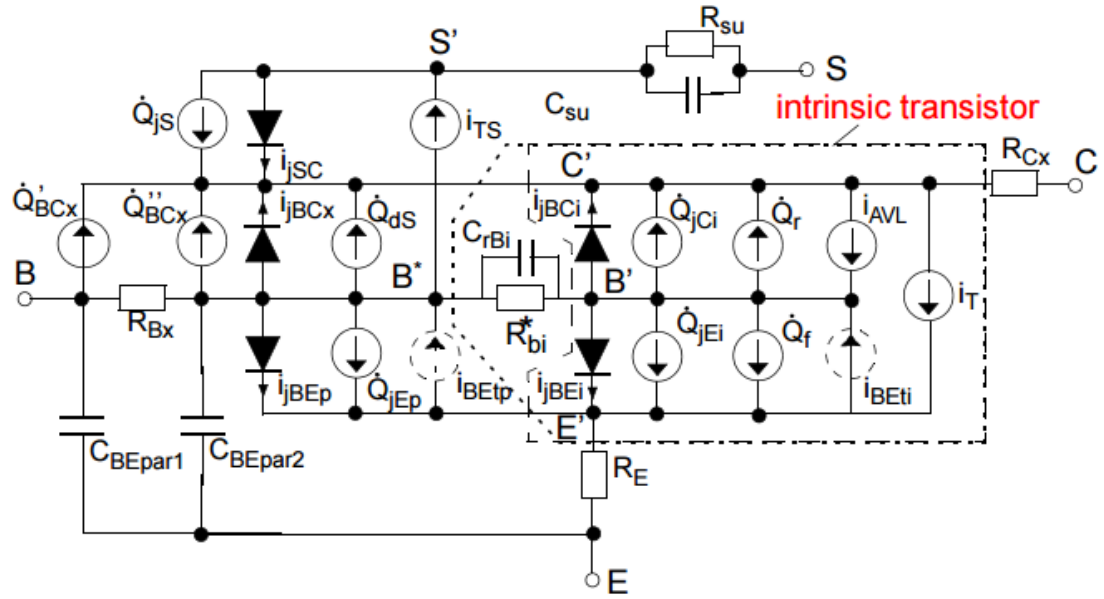
Modelul HICUM

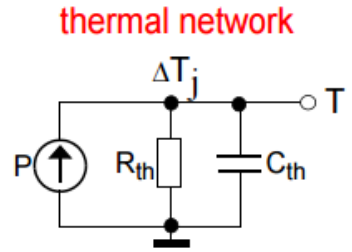
(HBT type)

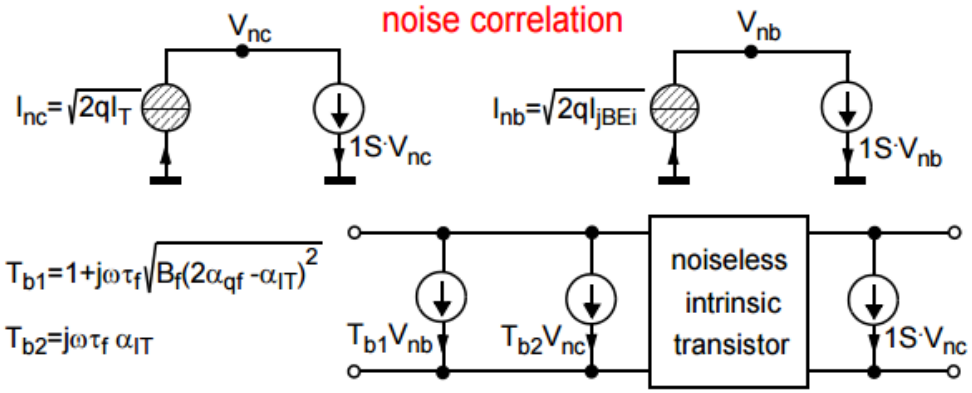


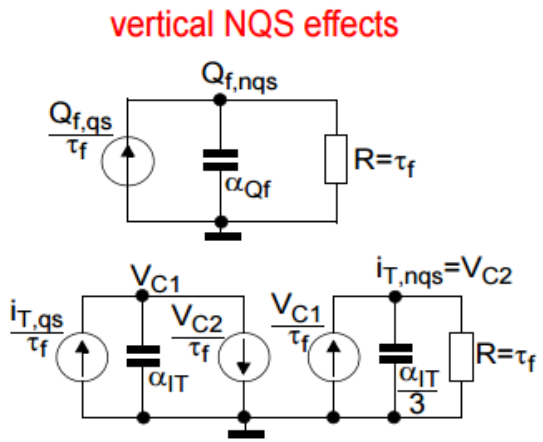
© https://www.iee.et.tu-dresden.de/iee/eb/forsch/Models/workshop0512/contr_2012/Schroter_HICUM_Ovw_slides.pdf

HICUM/L2 complete equivalent circuit



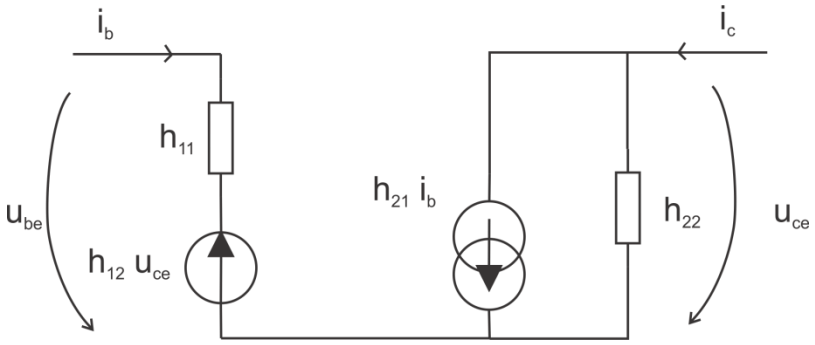






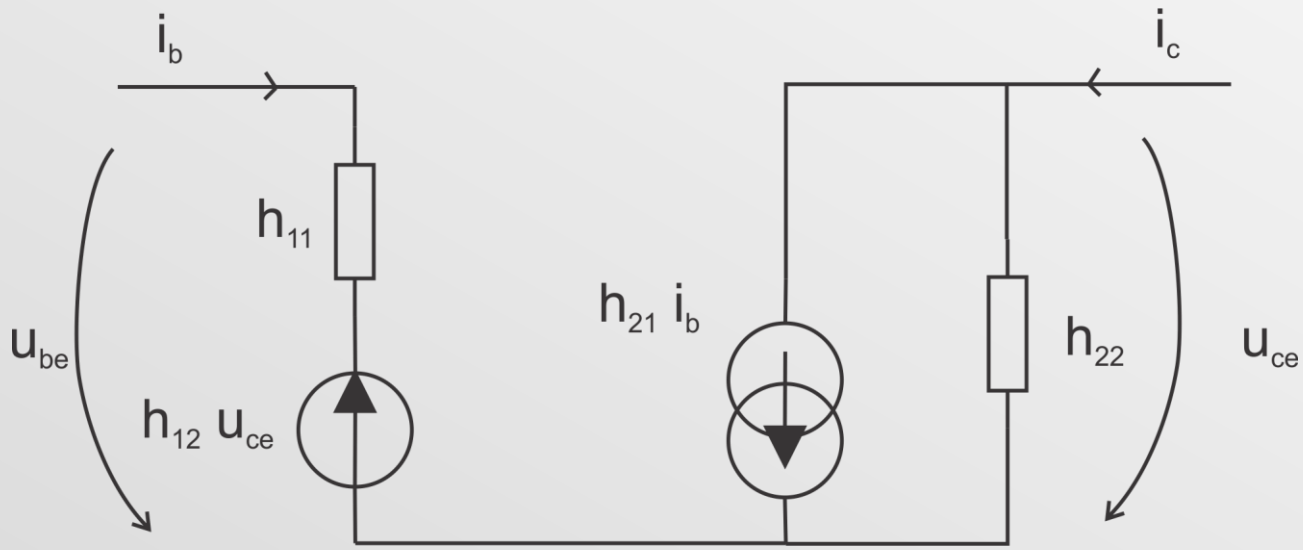
© https://www.iee.et.tu-dresden.de/iee/eb/forsch/Models/workshop0512/contr_2012/Schroter_HICUM_Ovw_slides.pdf

Parametrii hibrizi

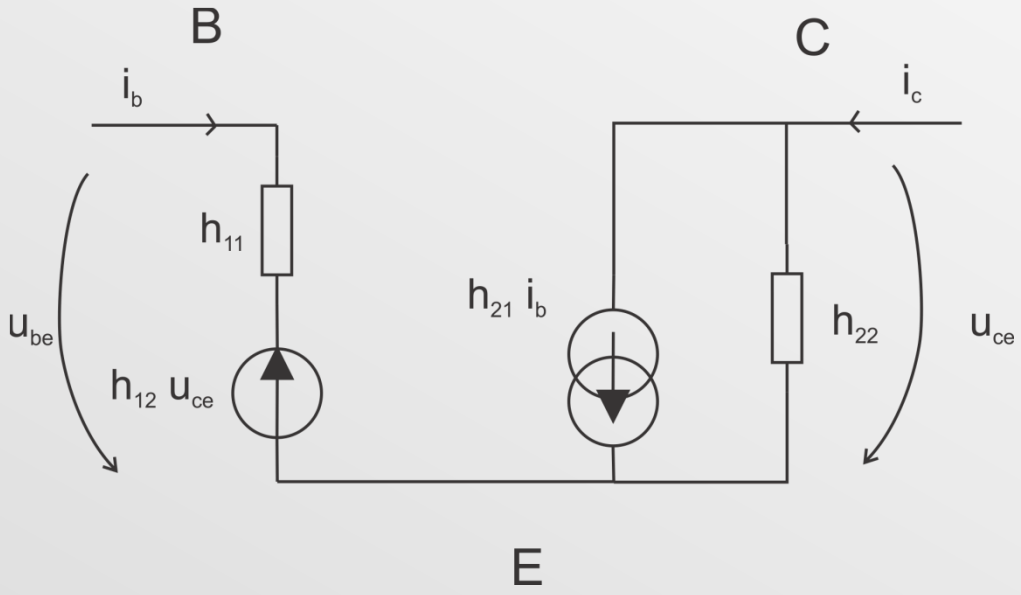


(revenim la BJT)

Modelul complet



Modelul complet



$$\begin{pmatrix} u_{be} \\ i_c \end{pmatrix} = \begin{pmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{pmatrix} \begin{pmatrix} i_b \\ u_{ce} \end{pmatrix}$$

$$u_{be} = h_{11} i_b + h_{12} u_{ce}$$

$$i_c = h_{21} i_b + h_{22} u_{ce}$$

Semnificații

$$u_{be} = h_{11} i_b + h_{12} u_{ce}$$

$$i_c = h_{21} i_b + h_{22} u_{ce}$$

h_{11} *leaga* u_i (u_{be}) *de* $i_i(i_b)$ => h_{input}

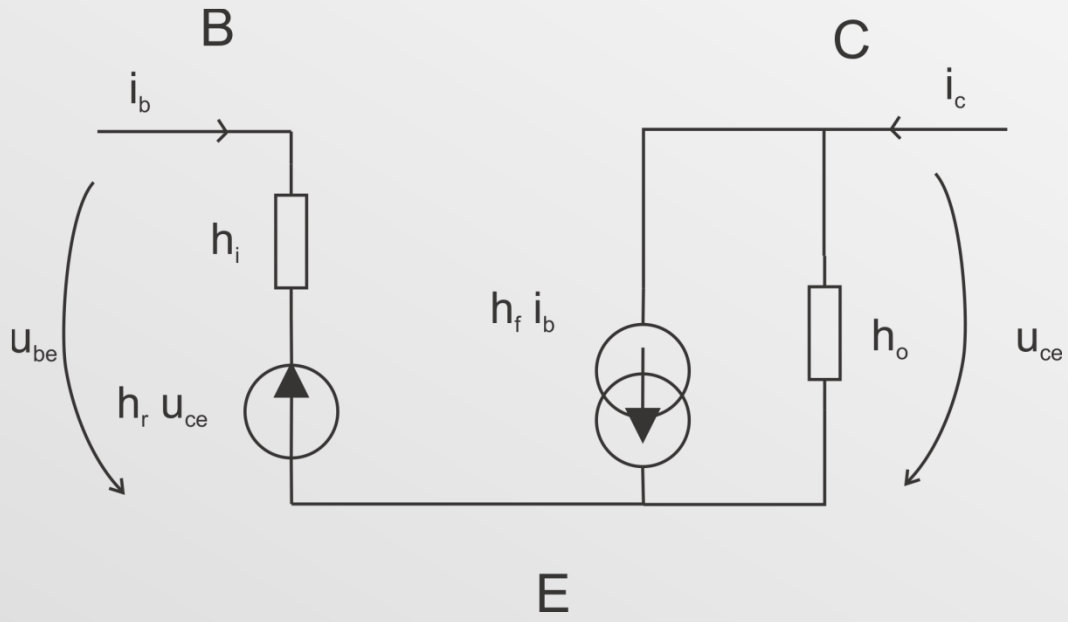
h_{21} *leaga* $i_o(i_c)$ *de* $i_i(i_b)$ => $h_{forward}$

h_{12} *leaga* $u_i(u_{be})$ *de* $u_o(u_{ce})$ => $h_{reverse}$

h_{22} *leaga* $i_i(i_c)$ *de* $u_o(u_{ce})$ => h_{output}

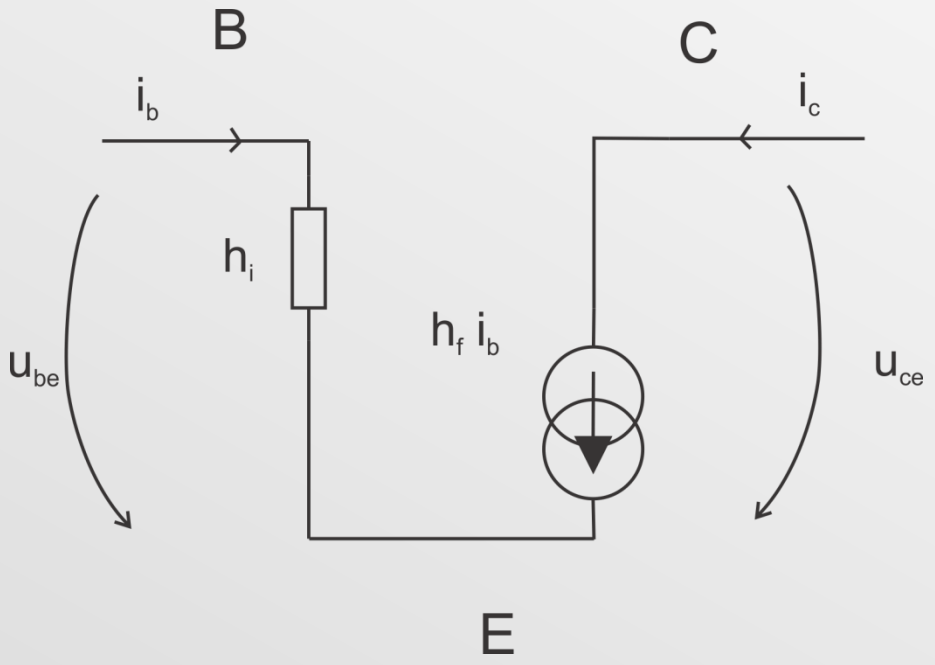
$$\begin{pmatrix} u_{be} \\ i_c \end{pmatrix} = \begin{pmatrix} h_i & h_r \\ h_f & h_o \end{pmatrix} \begin{pmatrix} i_b \\ u_{ce} \end{pmatrix}$$

Modelul complet



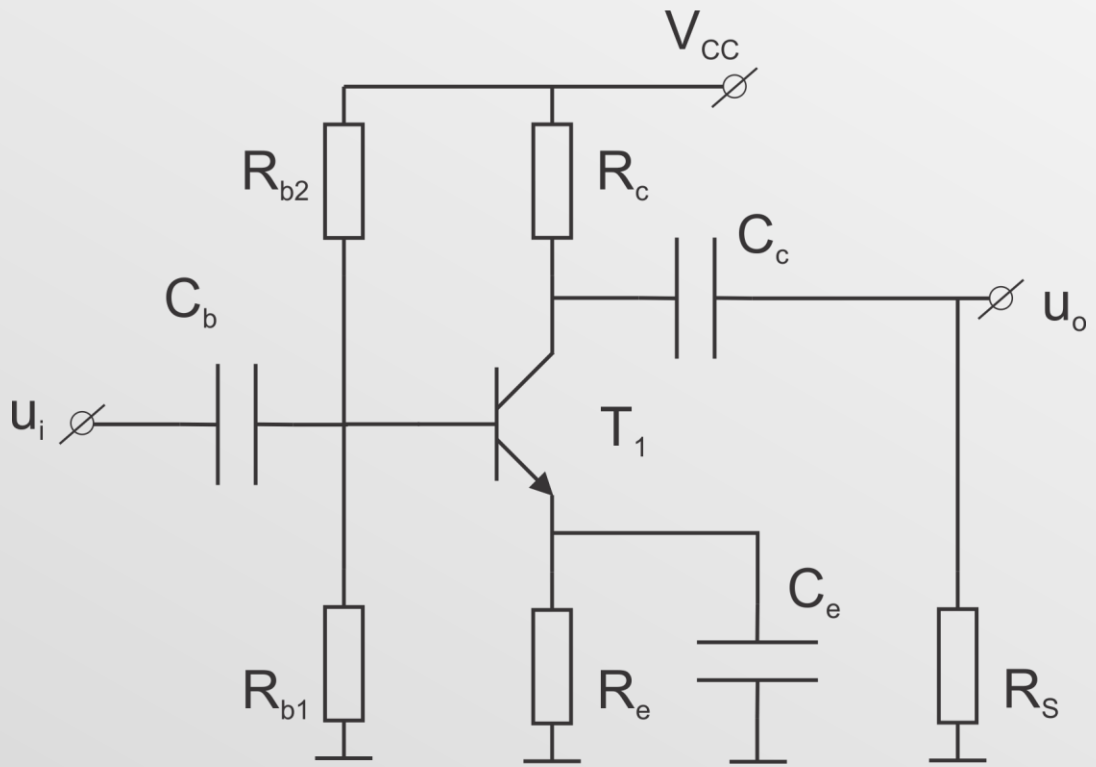
- h_i – Input impedance
- h_r – tipic - de ordinul 10^{-4} => se poate neglija
- h_f – Small signal current gain
- h_o – de ordinul μΩ => impedanță de ordinul MΩ => se poate neglija

Modelul simplificat

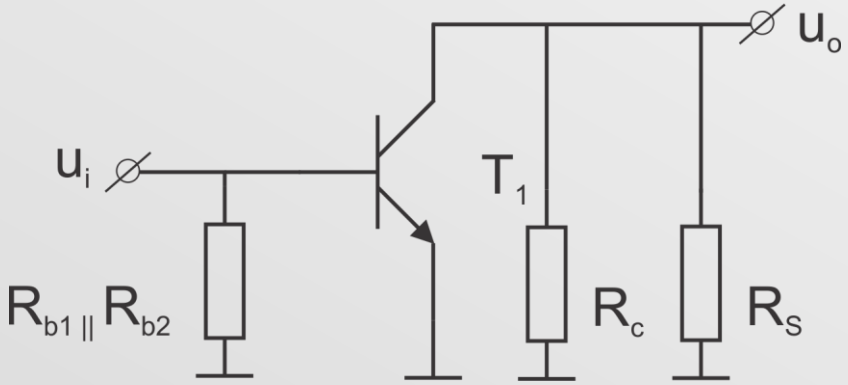


Conexiuni fundamentale

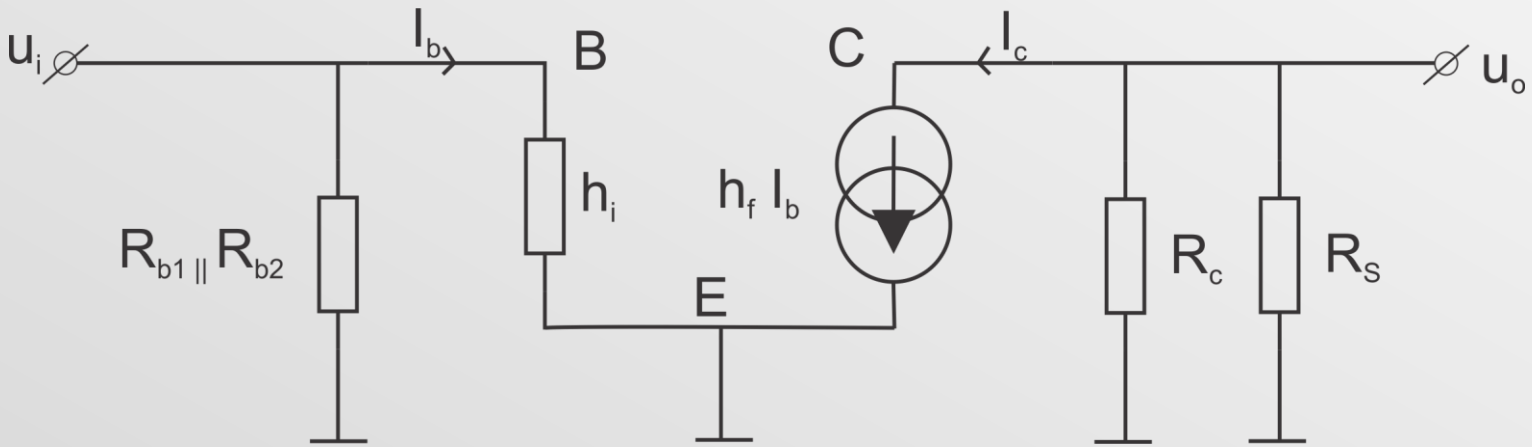
EC

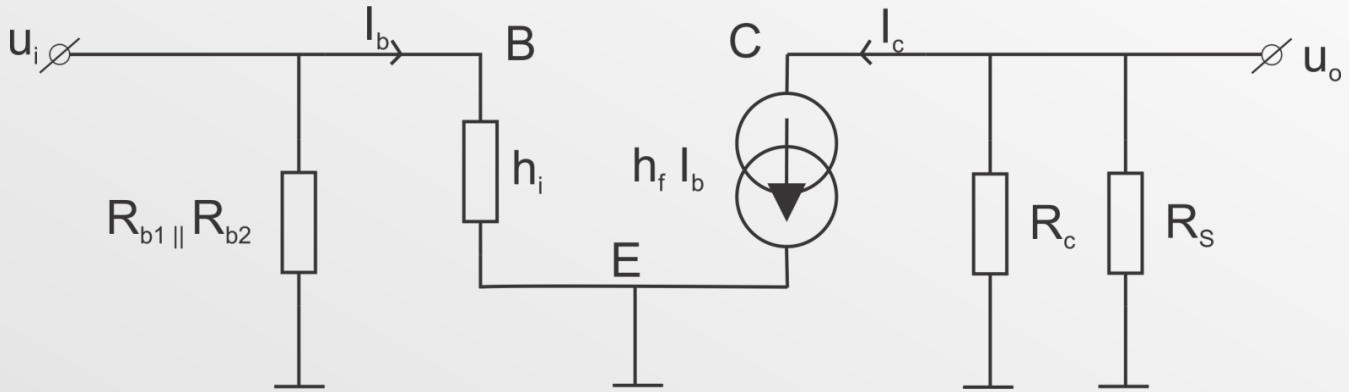


EC – regim dinamic



EC – param h





$$i_b = \frac{u_i}{h_i} \qquad i_c = h_f i_b = u_i \frac{h_f}{h_i} \qquad u_o = -i(R_c||R_s) = -u_i \frac{h_f}{h_i} R_c||R_s$$

$$A_u = -\frac{h_f}{h_i} R_c||R_s$$

$$Z_i = h_i \,||\, R_{b1} \,||R_{b2}$$

$$Z_o = R_c$$

$$A_i = \frac{i_o}{i_i} = \frac{u_o}{R_s} \frac{Z_i}{u_i}$$

Obs – intrinsec:

$$A_u = -\frac{h_f \; R_s}{h_i} = -S \; Z_s \qquad S \cong 40 I_C$$

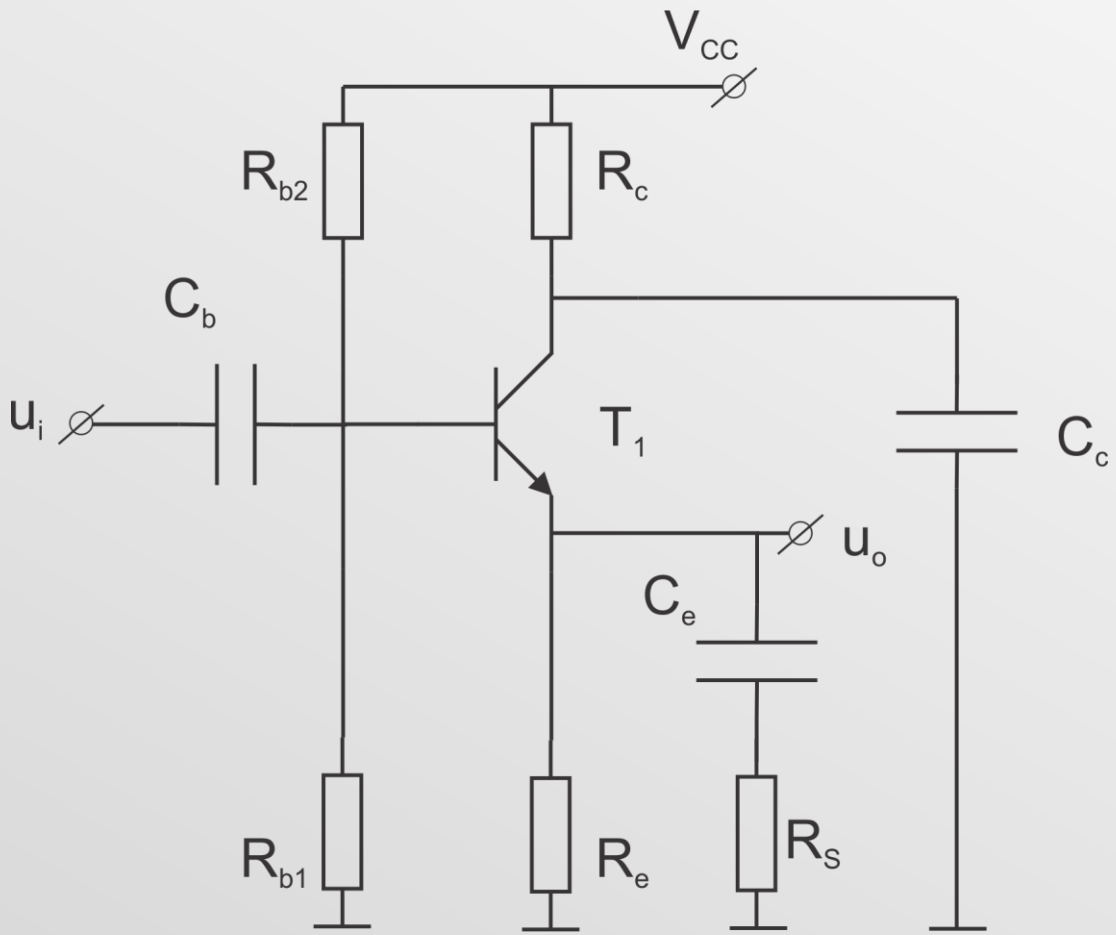
$$Z_i = h_i$$

EC - caracteristici

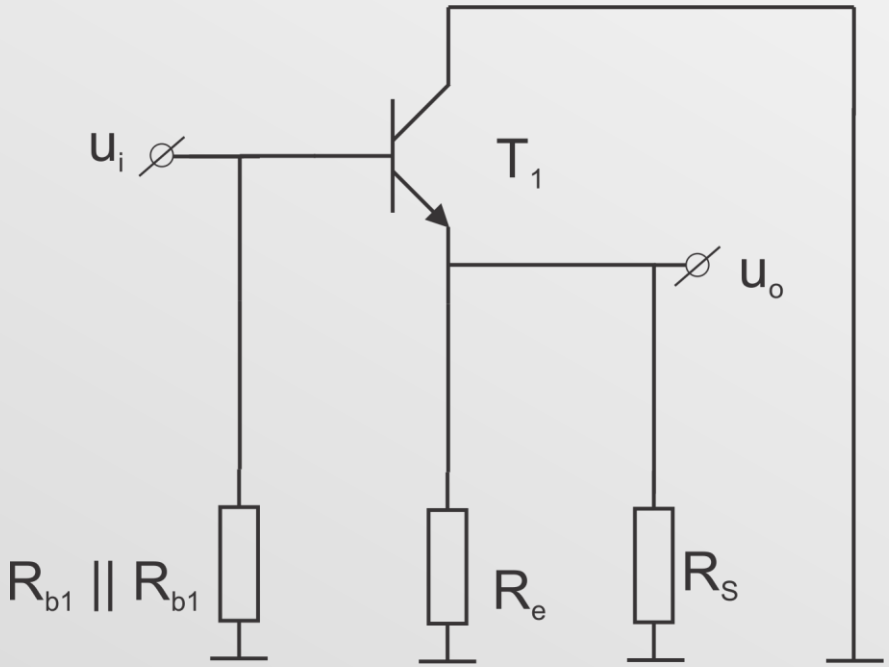
- Au
  - mare;
  - negativă (circuitul schimbă faza semnalului de la ieşire cu 180º faţă de faza semnalului de la intrare)
  - depinde de sarcină; Ordin: zeci – sute
  - depinde de Ic (de PSF)
- Ai - mare
- Ap - foarte mare
- Zi - mediu (prea mic) (sute Ω / kΩ)
- Zo - mediu (prea mare)

(doar intermediar)

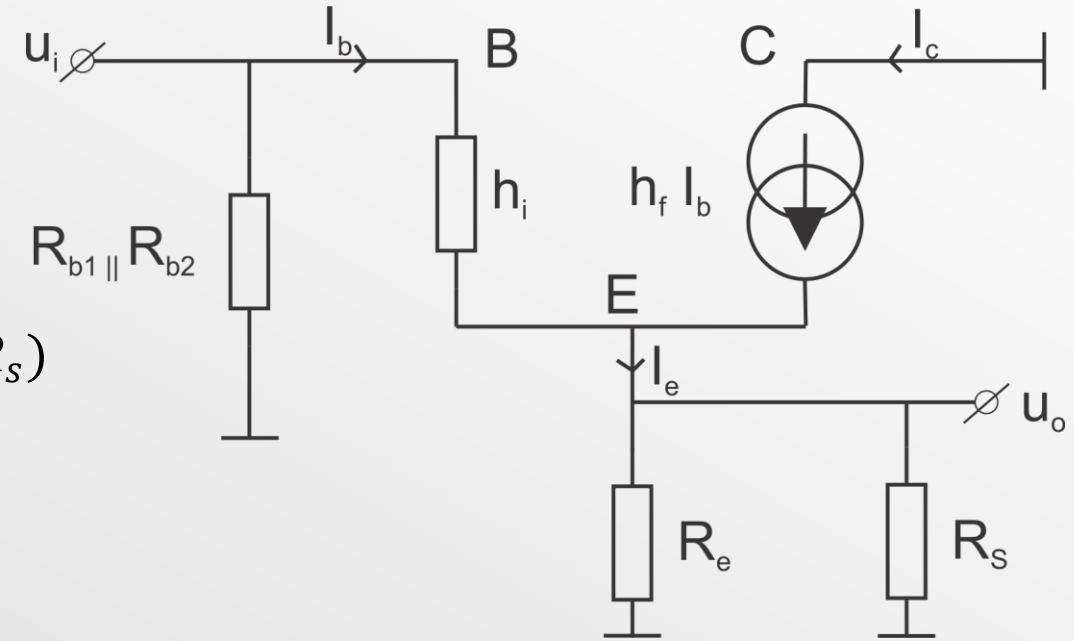
CC – Colector Comun



CC – regim dinamic



CC – param H



$$u_i = i_b h_i + u_o \qquad u_o = i_e (R_e || R_s)$$

$$i_e \cong i_c = h_f i_b$$

$$u_i = \frac{i_c}{h_f} h_i + i_c (R_e || R_s)$$

$$i_c = \frac{u_i}{(R_e || R_s) + \frac{h_i}{h_f}} \cong \frac{u_i}{(R_e || R_s)}$$

$$u_o \cong u_i$$

$$A_u \cong 1$$

$$Z_i = R_b || (h_i + (R_e || R_s) h_f)$$

$$Z_o = R_e || (\frac{R_b + h_i}{h_f})$$

$$A_i = \frac{i_o}{i_i} = \frac{u_o}{R_s} \frac{Z_i}{u_i}$$

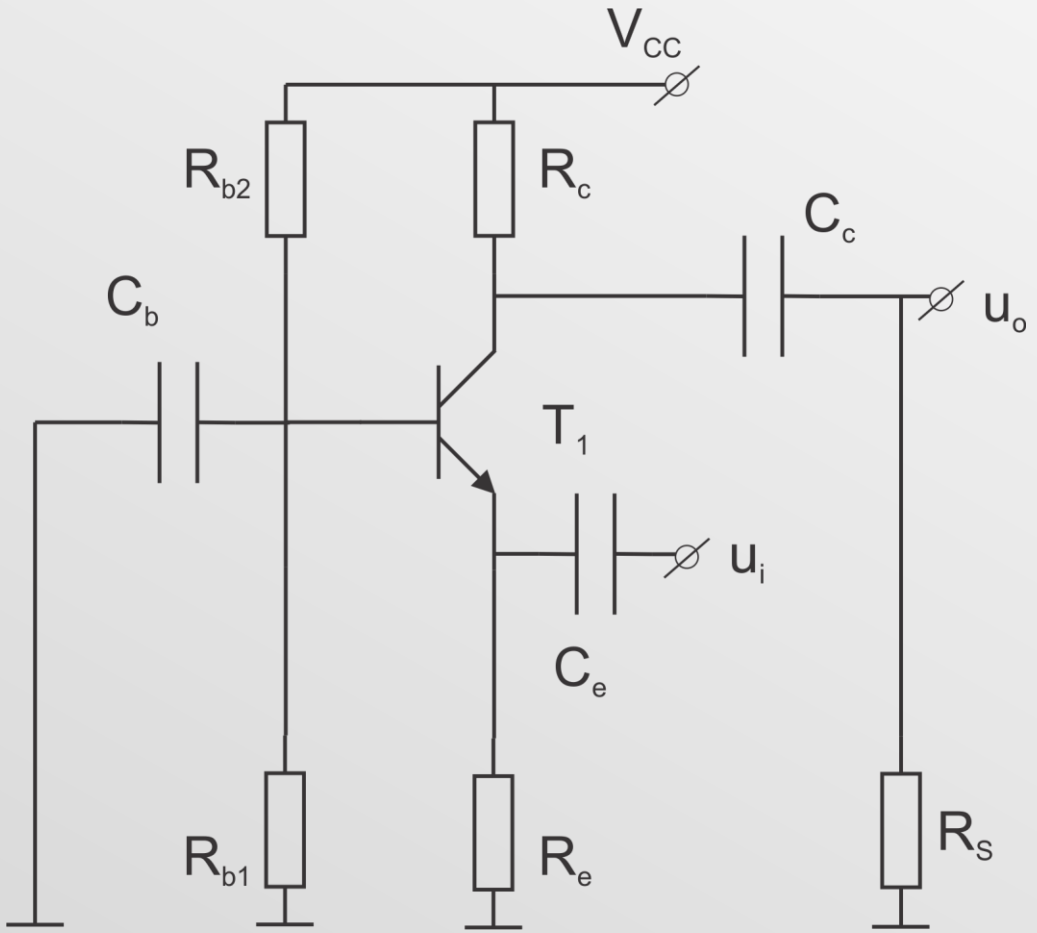
CC - caracteristici

- Au – subunitară, în general 1
- Ai - mare
- Ap - medie
- Zi – mare (zeci de kΩ)
- Zo - mic

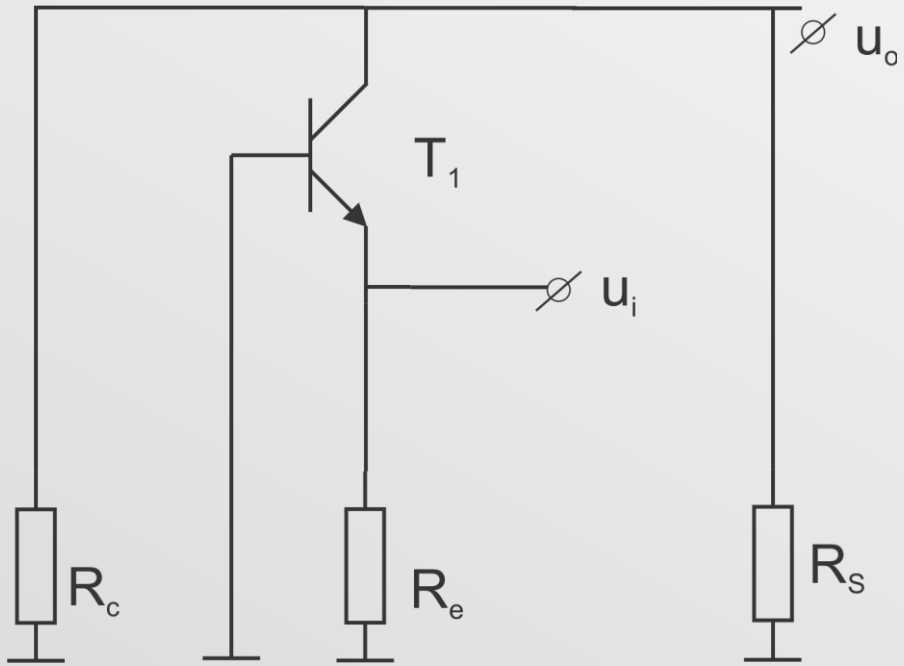
(bun pentru intrare / ieșire)

repetorul pe emitor este un foarte bun ***etaj de adaptare***

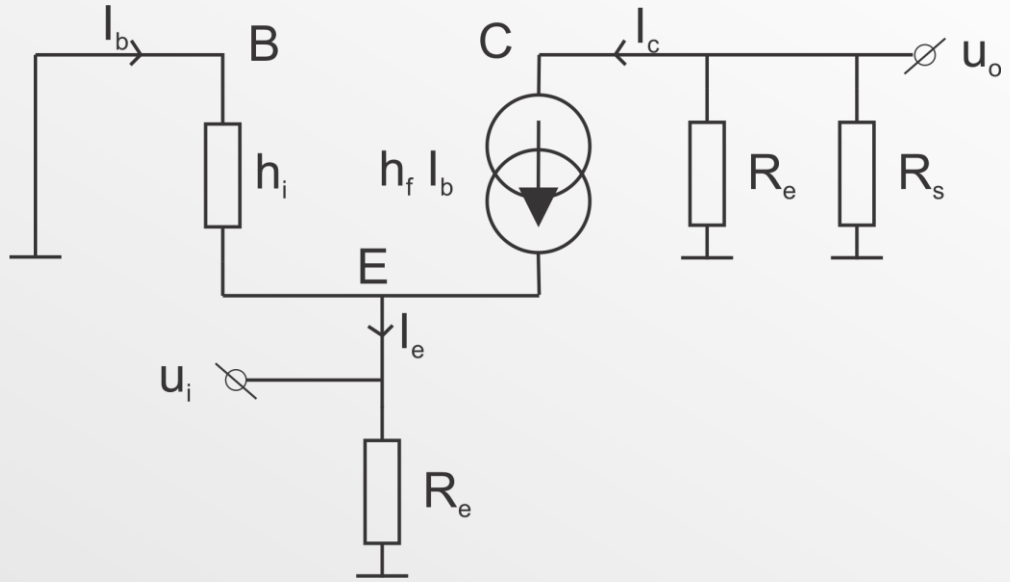
BC – baza comuna



BC – regim dinamic



BC – param h



$$i_b = -\frac{u_i}{h_i}$$

$$i_c = -\frac{u_i}{h_i}\, h_f$$

$$u_o = -i_c(R_c||R_s) = \frac{u_i}{h_i}\,(R_c||R_s)h_f = u_i S\,(R_c||R_s)$$

$$A_u = \frac{(R_c||R_s)h_f}{h_i}$$

$$Z_o = R_c$$

$$Z_i = R_e||\frac{h_i}{h_f} \cong \frac{h_i}{h_f}$$

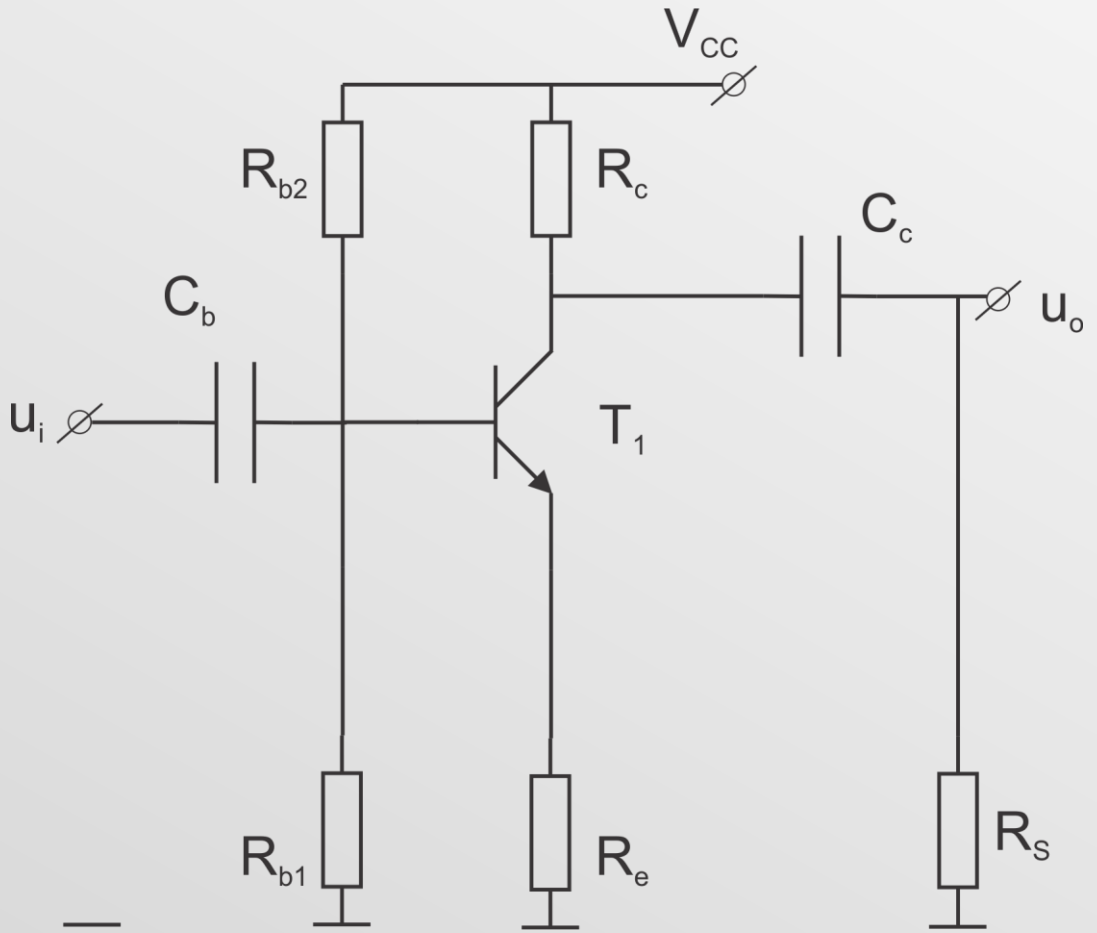
BC - caracteristici

- Au
  - mare;
  - pozitivă (= în modul cu EC)
  - depinde de Ic (de PSF)
- Ai – (-)1
- Zi – mică (zeci de Ω)
- Zo - mare

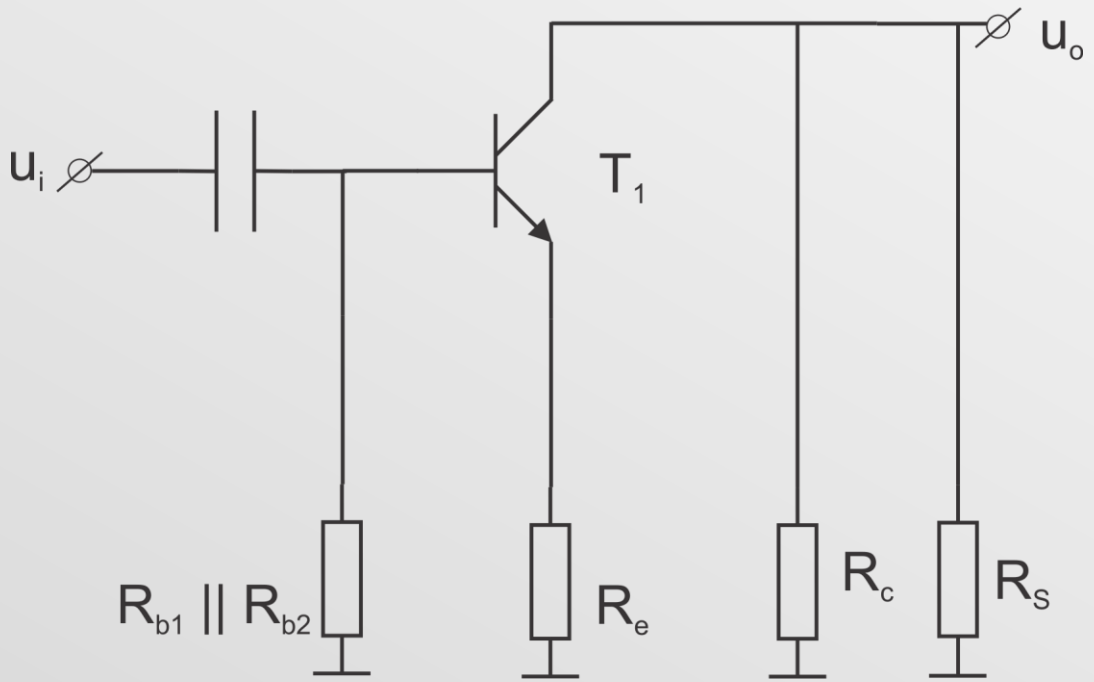
(bun pentru intrare în curent)

  - tranzistorul se comportă, la ieşire, ca un generator de curent şi în regim dinamic
  - scheme cu BM pot fi folosite ca sarcini dinamice sau ca generatoare de curent constant

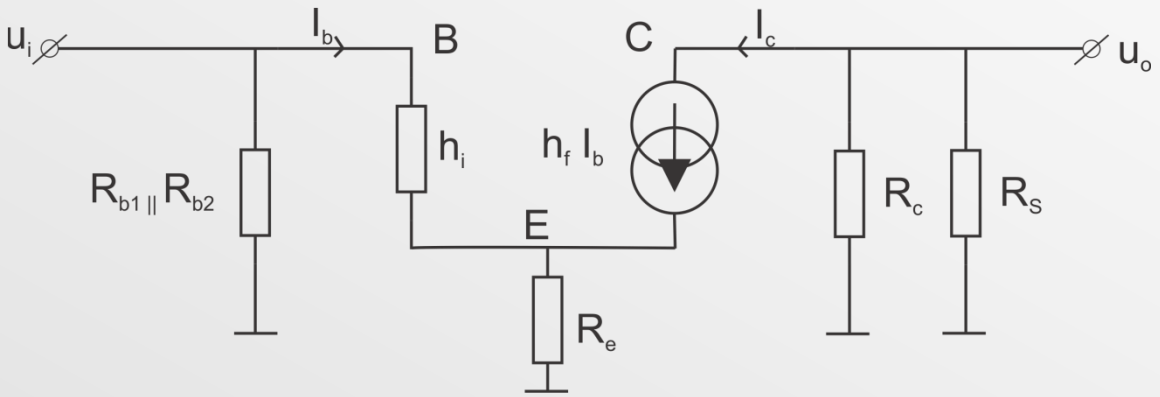
SD – sarcina distribuita



SD – regim dinamic



SD – param h



$$u_e \cong u_i$$

$$i_c \cong \frac{u_i}{R_e}$$

$$u_o = -i_c(R_c||R_s)$$

$$A_u \cong -\frac{R_c||R_s}{R_e}$$

$$Z_i = R_b||(h_i + R_e h_f)$$

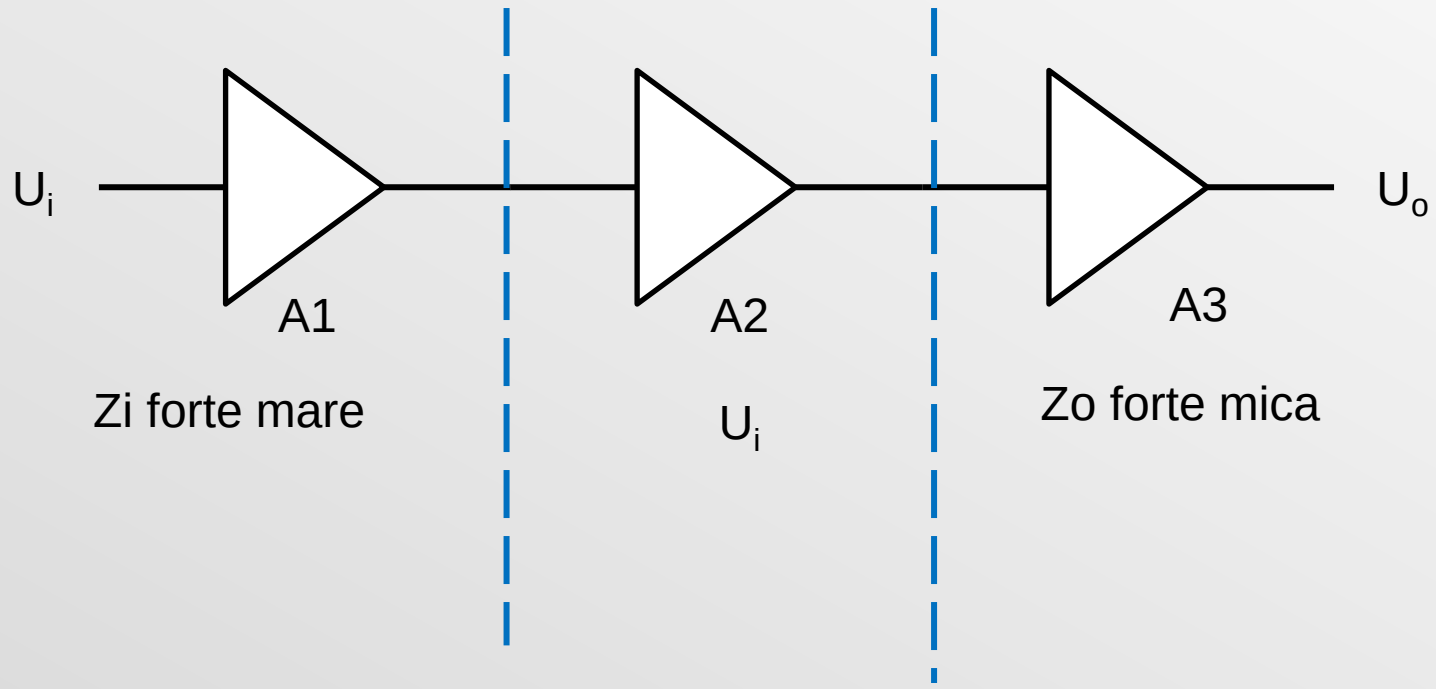
$$Z_o = R_c$$

SD - caracteristici

- Au – amplificare controlată bine
- Ai –
- Zi – mare (ok)
- Zo – mare (not ok)

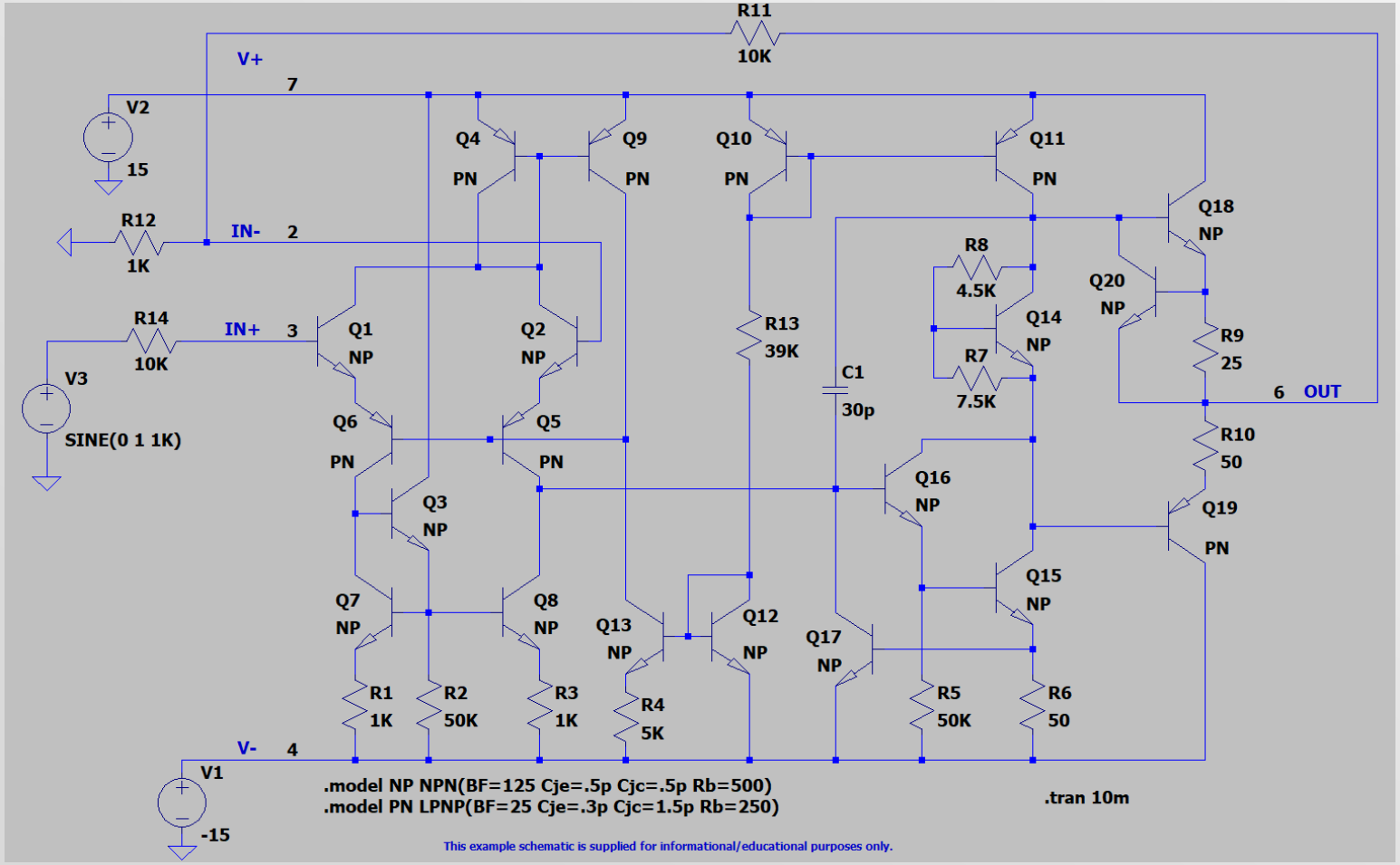
(util pentru circuite simple, cu amplificare fixă, cu un singur TBIP)

obs



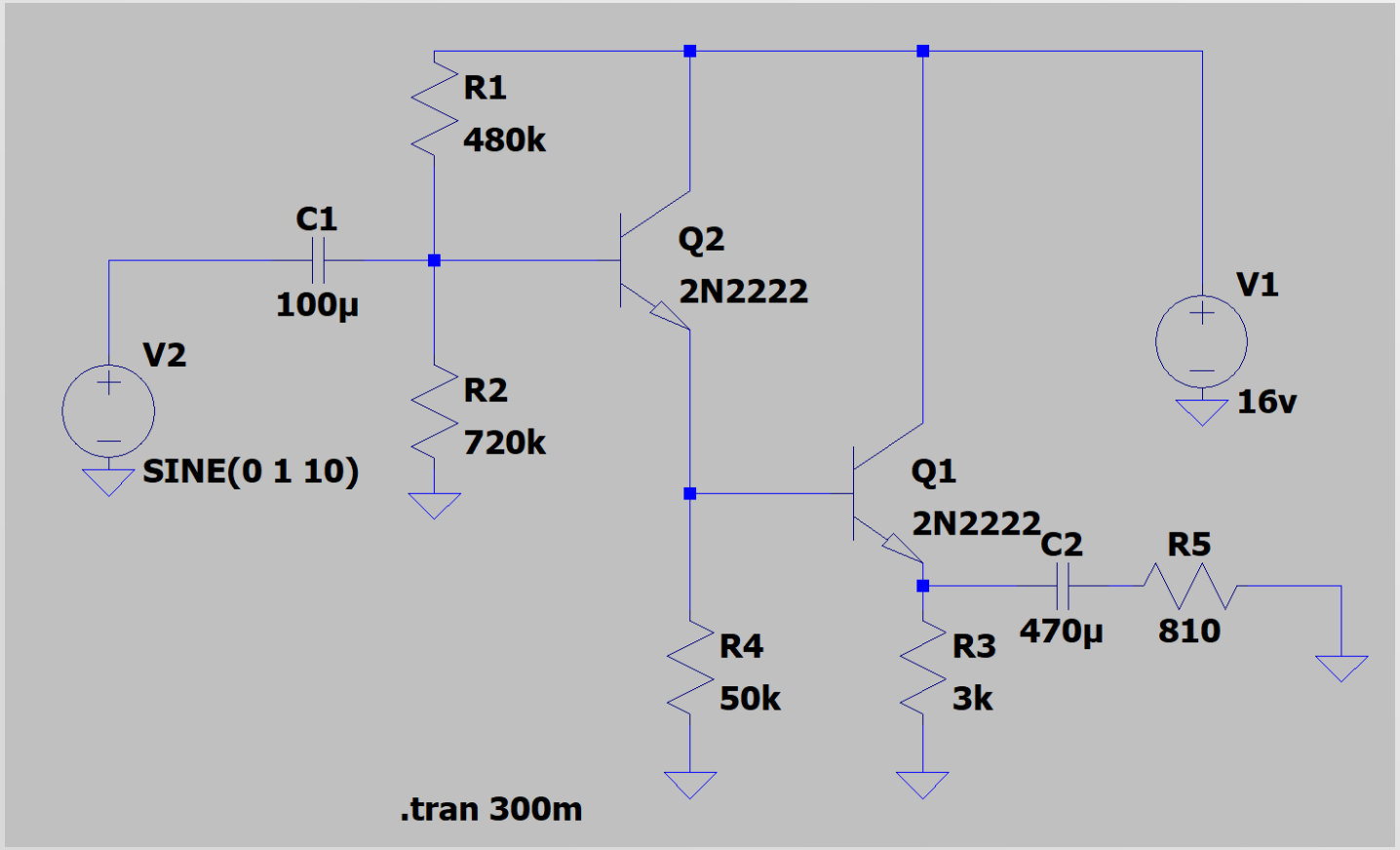
Zi = Zi1
Zo = Zo3
A = A1 A2 A3

obs


R11
10K
V+
7
V2
15
Q4
PN
Q9
PN
Q10
PN
Q11
PN
R12
1K
IN-
2
Q18
NP
R8
4.5K
Q20
NP
R14
10K
IN+
3
Q1
NP
Q2
NP
R13
39K
Q14
NP
R9
25
C1
30p
R7
7.5K
6
OUT
V3
SINE(0 1 1K)
Q6
PN
Q5
PN
R10
50
Q16
NP
Q19
PN
Q3
NP
Q15
NP
Q7
NP
Q8
NP
Q13
NP
Q12
NP
Q17
NP
R1
1K
R2
50K
R3
1K
R4
5K
R5
50K
R6
50
V-
4
V1
-15
.model NP NPN(BF=125 Cje=.5p Cjc=.5p Rb=500)
.model PN LPNP(BF=25 Cje=.3p Cjc=1.5p Rb=250)
.tran 10m
This example schematic is supplied for informational/educational purposes only.

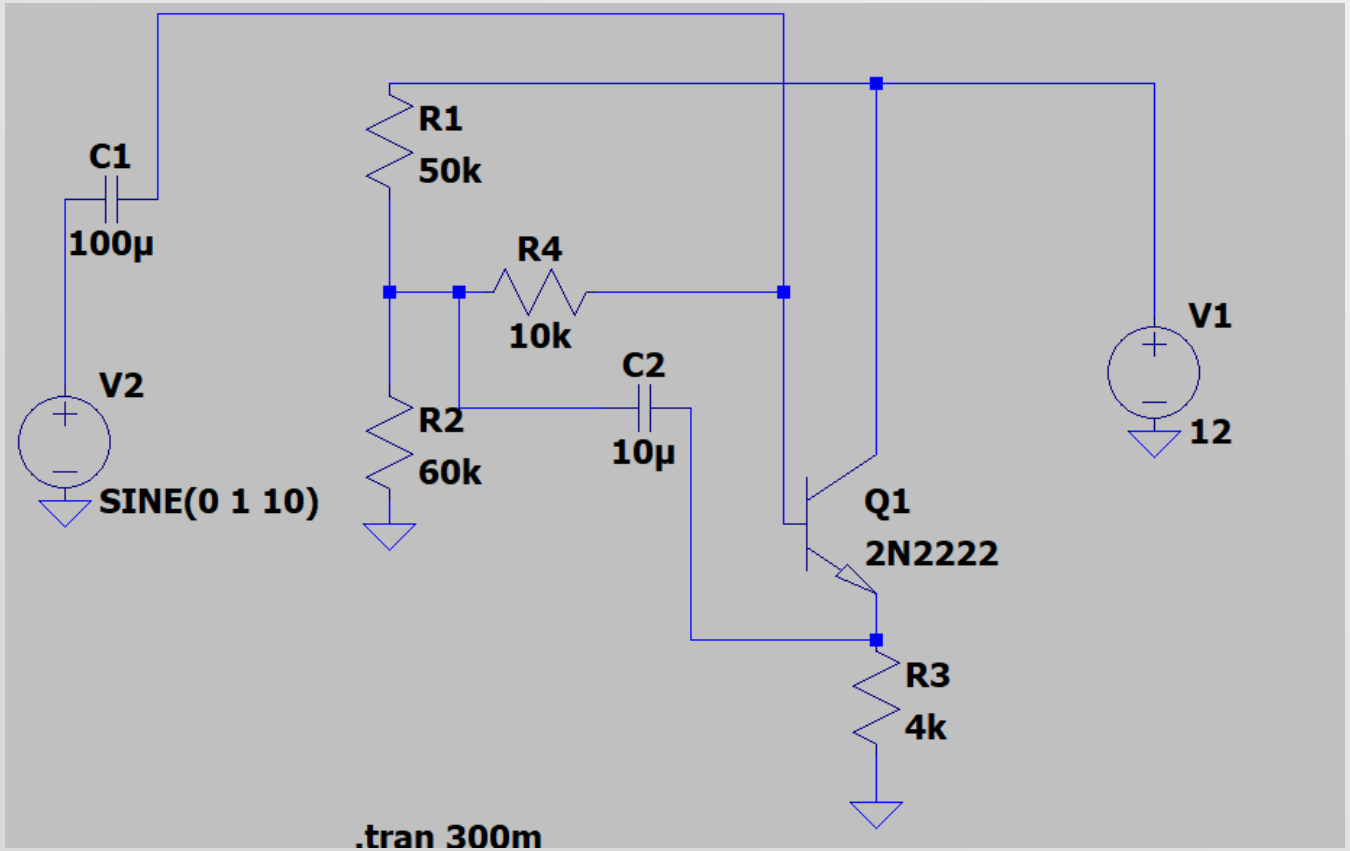
Darlington

- (la curs)



Bootstrap (just for fun, no exam)

- (la curs)



Discutie aplicatii D

- (la curs)

https://www.ti.com/lit/ds/symlink/uln2003a.pdf

https://tinyurl.com/yy45yr5d